

Electric Grid Fragility Analysis

An Applied Analysis of Grid Stress Using Public Operational Data

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Full Documentation and codebase available here -
https://github.com/miles-wolf/Electric_Grid_Fragility_Analysis

1. Executive Summary

This report presents a framework for measuring and comparing operational stress across electric grids using publicly available EIA-930 hourly data from 2025.¹ The analysis covers two balancing authorities (BAs) - CAISO in California, and PJM on the East Coast - and focuses on determining how hard each system is being asked to work relative to its own norms.

A normalized fragility score is built from three demand-side indicators: peak load, maximum ramp rate, and the fraction of hours spent near peak. Scores are standardized within each BA, so results are comparable across systems of different sizes. A data quality filter is applied at the hourly level before any calculations to remove reporting artifacts. Machine learning analysis is applied to determine the trends in fragility data and classify operating regimes (Figure 1).

Key findings:

- Fragility is episodic. Most days are ordinary. Stress concentrates in a small number of events.
- CAISO stress is driven by peak load and ramp magnitude occurring together, tied to late-summer heat events that intensify the solar duck curve.
- PJM stress is driven primarily by sustained high load at scale and less by daily fluctuations
- These patterns hold across three independent analyses: cluster geometry, feature contribution on high-fragility days, and temporal event trajectories

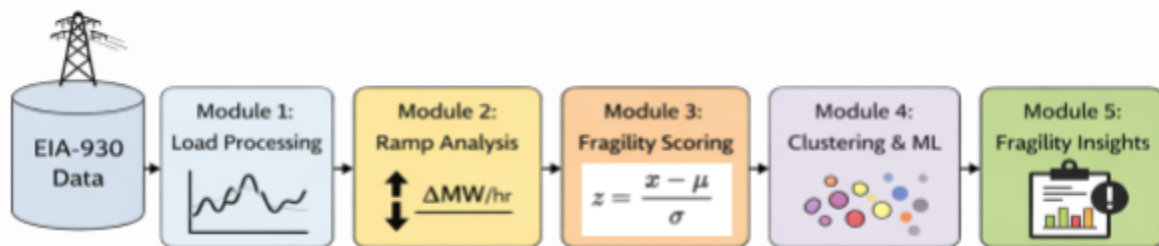


Figure 1 - Grid Fragility Analysis Process Flowchart

This flowchart illustrates the full analysis workflow, showing how raw EIA-930 demand data is transformed via successive modules into fragility scores and downstream modeling inputs.

2. Data and Scope

This analysis draws on EIA Form 930 consolidated hourly operational data for two balancing authorities - CAISO and PJM Interconnection - covering the full calendar year 2025 at hourly resolution, aggregated to daily features for modeling. Data was filtered as follows. Single-hour demand readings deviating from their 3-hour rolling median by more than 25% of that day's load range are flagged as reporting artifacts. Flagged readings are replaced by linear interpolation from adjacent hours. The filter is self-calibrating - no BA-specific thresholds required. One reading was corrected in CAISO (7/31/25), where a corrupted value would have generated phantom ramp events large enough to consume an entire K-Means cluster (see Appendix A).

3. Measuring Grid Fragility

Z-scores are calculated for three separate categories - daily peak load Z_{DPL} , daily maximum hourly ramp Z_{MHR} , and fraction of hours near daily peak Z_{HNP} (equation 1). Zero means a typical day. The within-BA standardization is what makes cross-system comparison valid even though their absolute load scales may differ significantly. The three Z-scores were then added using equal weights to obtain an overall fragility score (equation 2). Lastly, fragility scores were normalized to a 0-1 scale so they could be easily cross analyzed between BAs (equation 3). Results show similar, not identical, annual trends for CAISO and PJM (Figure 2) (see Appendix A for feature engineering and weighting assumption details).

$$Z = (X - \mu) / \sigma \quad \text{where } \mu \text{ is the annual mean and } \sigma \text{ is the standard deviation} \quad (\text{equation 1})$$

$$Z = Z_{DPL} + Z_{MHR} + Z_{HNP} \quad (\text{equation 2})$$

$$Z_{norm} = (Z - Z_{min}) / (Z_{max} - Z_{min}) \quad (\text{equation 3})$$

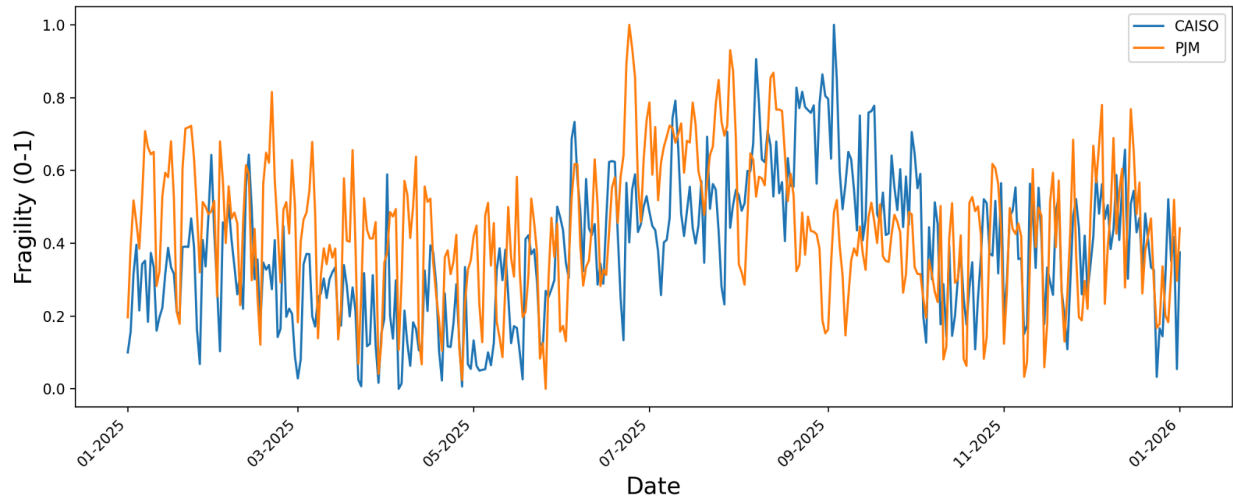


Figure 2. Daily Grid Fragility Time Series

Daily fragility for CAISO and PJM over calendar year 2025. Both grids peak in summer; CAISO is lower in winter and more concentrated seasonally.

4. Operational Regimes & Stress Mechanisms

K-Means clustering on PCA-reduced features identifies four repeatable operating regimes in each BA. The regime geometry is similar between systems, though the stress drivers differ. On CAISO's high-fragility days, peak load and max ramp contribute nearly equally (42% and 43%), with limited near-peak persistence (15%). These are California late-summer heat events, where high afternoon demand and steep evening solar ramps hit at the same time. PJM's high-stress

days are more load-led (45% peak, 34% ramp), with near-peak persistence playing a larger role (21%) (Figure 3). These are Eastern heat dome events — sustained high demand across a large footprint (see Appendix C). Both systems' high-stress regimes sit in the upper-right of feature space, combining elevated peak load and elevated ramp simultaneously — tracking roughly along the $y=x$ diagonal rather than separating along either axis alone (Figure 4).

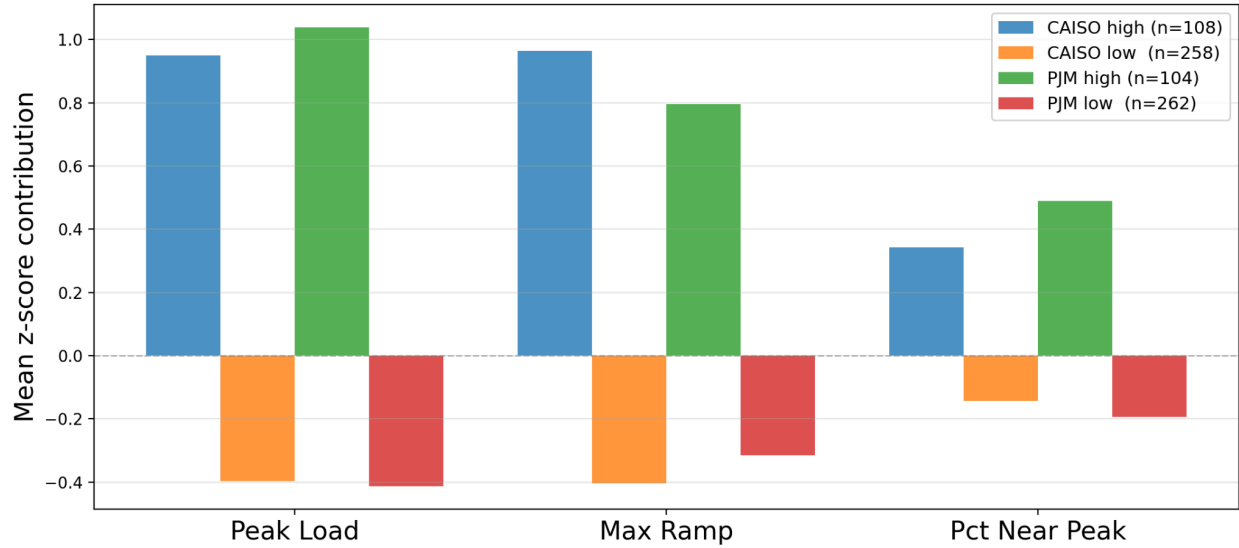


Figure 3. Mean z-score contribution of each feature

Data split into high vs. low fragility days for CAISO and PJM. High-fragility days are defined as composite fragility_z > 1.0 (CAISO n=108, PJM n=104); all other days form the low group. Bar height reflects the average z-score for that feature across days in each tier.

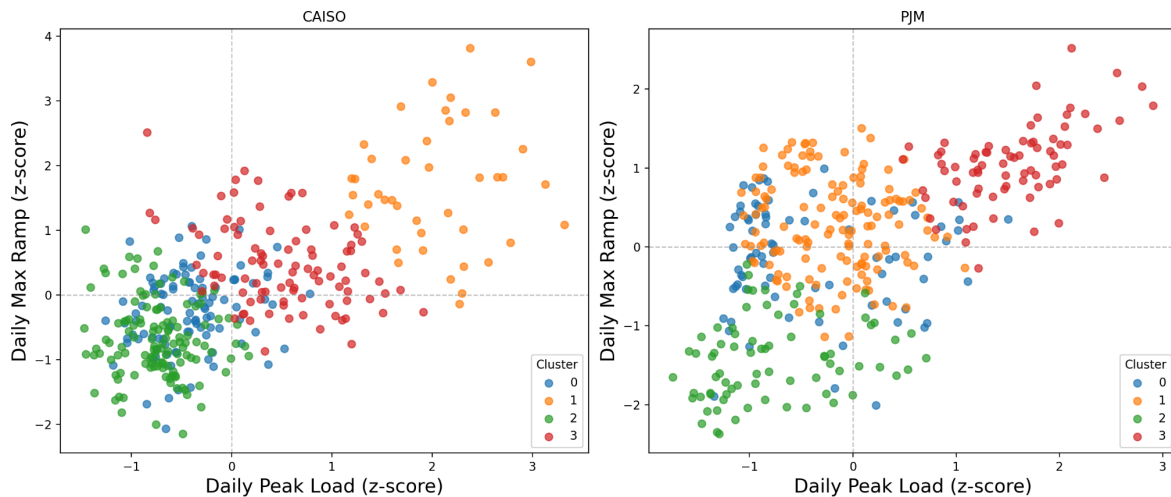


Figure 4. Operational regimes in normalized feature space

CAISO and PJM operating regimes in normalized feature space (peak load z-score vs. max ramp z-score). PJM's high-stress regime (Cluster 3) is more load-driven, while CAISO's high-stress regime (Cluster 1) combines peak load and ramp stress more equally.

Both systems show a four-tier progression from baseline to high stress, with the low and high ends well-separated from the middle. In CAISO, the two moderate clusters (0 and 3) sit nearly on top of each other in mean fragility — distinguishable in feature space but nearly identical in overall stress level, suggesting the clustering is capturing operational differences that the composite score alone doesn't fully resolve. CAISO's high-stress cluster has a notably wide spread, reflecting real variability among summer peak events. PJM's progression is cleaner, with each tier evenly spaced, though its high-stress cluster contains a notable outlier (Figure 5). Regime frequency reinforces the key structural difference: CAISO's high-stress regime occurs on only ~12% of days, while PJM's occurs on ~21%. CAISO stress is more concentrated in time; PJM accumulates it more steadily (Figure 6) (see Appendix B, C for clustering details).

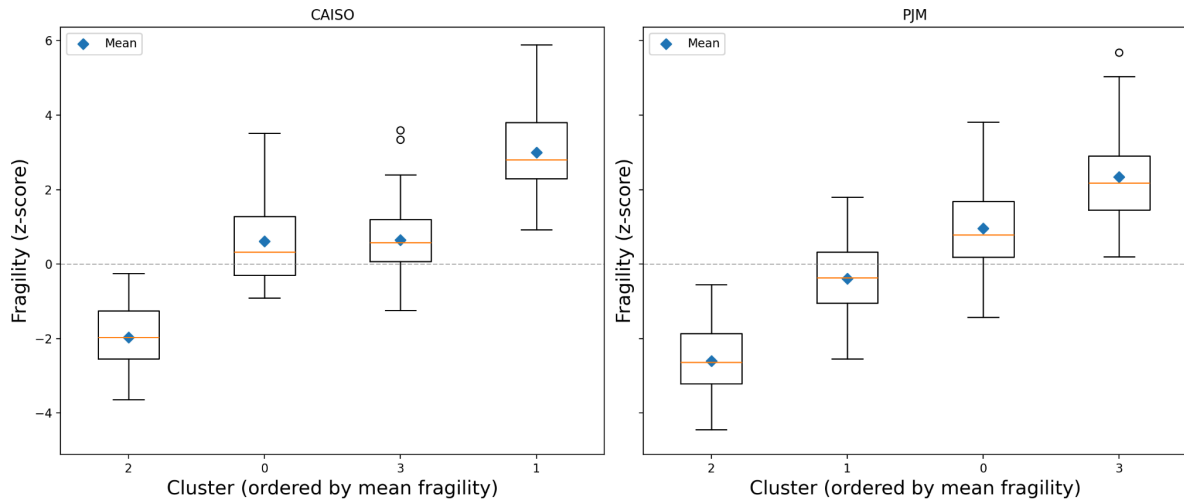


Figure 5. Fragility vs cluster box plot

Fragility distributions by cluster for each BA, ordered by mean fragility (before normalization).

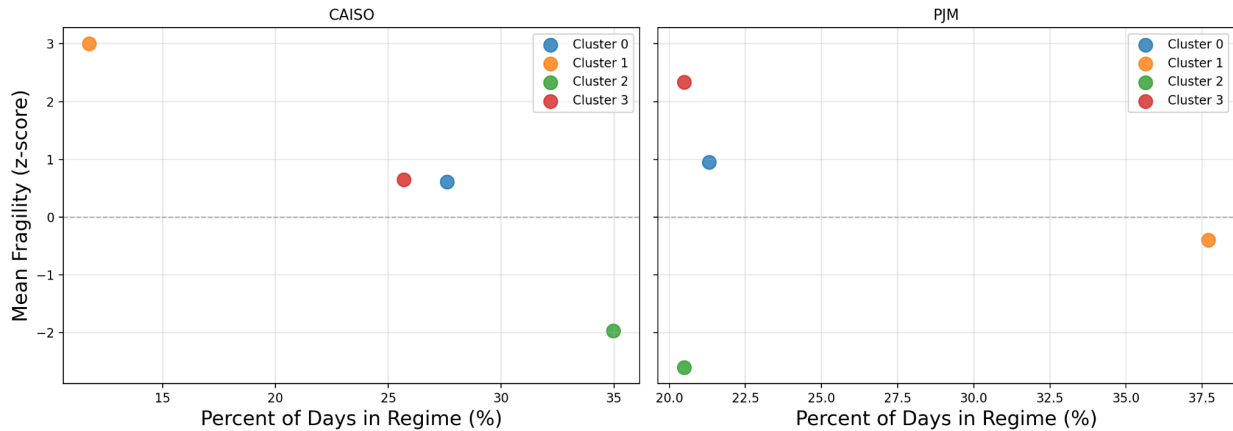


Figure 6. Regime frequency versus mean fragility

Mean fragility score for each regime versus share of days spent in each operational regime, for CAISO (left) and PJM (right). Each point represents one cluster. CAISO's highest-stress regime is rare (~12% of days) but acute; PJM's is more frequent (~21%) and more moderate.

5. Extreme Fragility Events

The top CAISO events (9/3, 8/7, 8/21) share a consistent structure: peak load and ramp rise together over several days, fragility builds to 5–6 z-score, then both fall off symmetrically. These are not isolated single-day spikes — the system spends multiple consecutive days in the high-stress regime before recovering. Notably, CAISO's ramp signal is more volatile than its load signal during these events — the max ramp line jumps day-to-day even as load rises more smoothly. This reflects the solar duck curve dynamic: afternoon ramps are being driven not just by demand growth but by the daily collapse of solar generation, which introduces a ramp component that is partially decoupled from load level. (Figure 7)

PJM's top events (7/24, 7/29, 8/13) look structurally different. Peak load builds slowly and symmetrically over roughly a week — the approach and recovery are nearly mirror images of each other. Ramp tracks load closely and at comparable amplitude, with none of the day-to-day volatility seen in CAISO. Fragility accumulates gradually, stays elevated for several consecutive days near the peak, and then recedes at the same pace it arrived. The system enters the high-stress cluster regime for extended stretches, not brief windows. This is the heat dome signature — a large, thermally-inertial system warming and cooling slowly across a wide geographic footprint, with no sharp solar-driven ramp component to add noise. (Figure 8)

Taken together, these trajectories reinforce the regime-level findings: CAISO stress is acute and ramp-amplified; PJM stress is sustained and load-led. The pattern holds consistently across all six major extreme events examined between the two BAs.

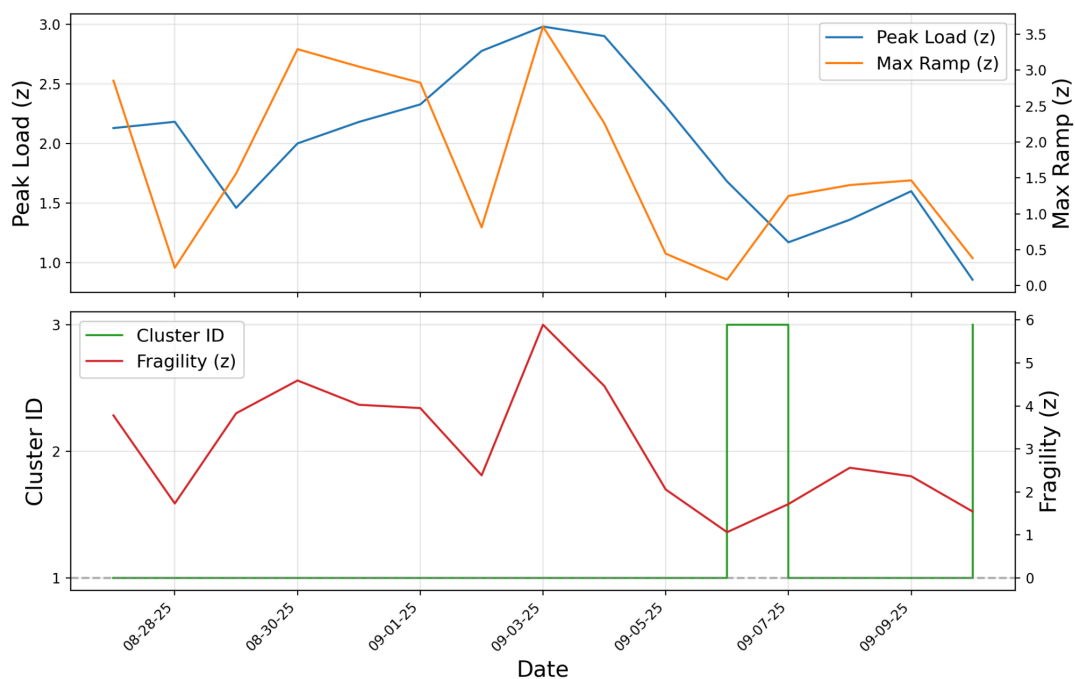


Figure 7. CAISO extreme fragility event — combined temporal trajectory
CAISO highest-fragility event (centered Sept 3, 2025). Peak load and ramp rise and fall together over several days; fragility peaks near 6 z-score.

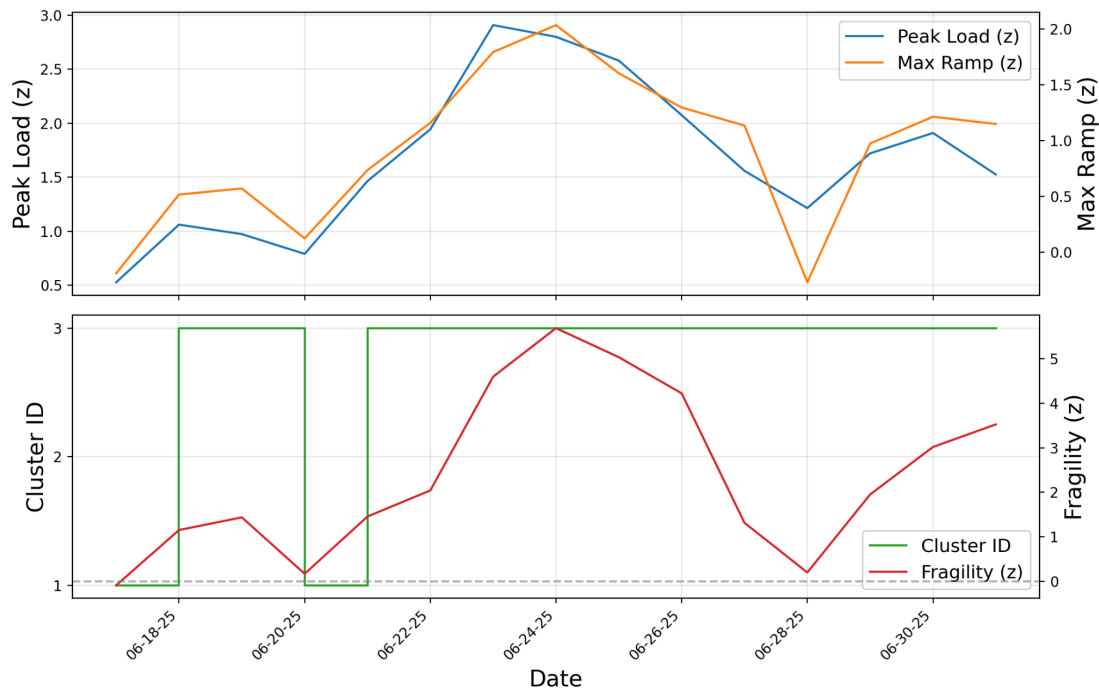


Figure 8. PJM extreme fragility event — combined temporal trajectory

PJM highest-fragility event (centered June 24, 2025). Peak load builds smoothly over a week; fragility accumulates gradually and sustains.

6. Next Steps

This framework characterizes what grid stress looks like and when it occurs — it doesn't explain why, and it doesn't predict. There's no supply-side information, so it can't distinguish between a day where fragility was high but managed easily and one where it strained reserves.

The natural extensions from here:

- More BAs. The code is self-calibrating and runs on any EIA-930 region. Adding ERCOT, MISO, NYISO, and SPP would test whether the CAISO/PJM archetypes generalize or if other stress profiles emerge.
- External linkage. Joining high-fragility days to weather records, electricity market prices, and NERC event logs would start to move from description toward understanding causes.
- Predictive modeling. The regime labels are clean, interpretable targets. Forecasting next-day regime probability from leading indicators (weather forecasts, prior-day fragility, seasonal features) is the most direct path toward an operational early-warning tool.

- Electric Grid Design: As grids evolve to accommodate large new loads — data centers, EV charging infrastructure — and distributed generation like utility-scale solar, fragility mapping can inform where and when new capacity additions create the most stress relief or the most risk. Siting a data center in a regime that already drives peak fragility looks very different from placing one in a low-stress baseline regime (see Appendix C for further discussion)

7. Conclusion

The framework delivers a clean, reproducible way to characterize grid operational stress from public data. It can describe where and when stress concentrates and identify patterns across systems, but it doesn't yet explain the underlying mechanisms or tell you what to do about them.

The CAISO and PJM findings are specific and consistent across multiple analyses. The pipeline scales to any balancing authority without modification. The next version of this work gets more interesting when supply-side data and external event records enter the picture in order to build towards implementing robust prediction mechanisms that can be used to make grid operational and design decisions.

Appendix

Appendix A: Fragility Score Construction and Normalization

All features are computed from hourly EIA-930 demand readings aggregated to the daily level, independently for each BA. Five daily features are derived before any normalization is applied.

- Daily Peak Load (MW): Max observed demand reading within the calendar day.
- Daily Minimum Load (MW): Min observed demand reading within the calendar day.
- Daily Load Range (MW): Difference between daily peak and daily minimum. Captures how much the system swings within a single day.
- Maximum Hourly Ramp (MW/hr): Largest absolute change in demand between any two consecutive hourly readings within the day. Computed as the maximum of the absolute first difference of the hourly demand series. Sign is dropped — both up-ramps and down-ramps count equally.
- Fraction of Hours Near Peak: Share of hours in a given day where demand was at or above 95% of that day's peak load. A value of 0.10 means 10% of hours — roughly 2–3 hours — were within 5% of the daily maximum. This feature captures load persistence: a day that peaks sharply and recovers quickly scores low, while a sustained heat event scores high. The 95% threshold is a fixed design parameter.

These five features are then z-scored within each BA before PCA and clustering. Daily Peak Load, Maximum Hourly Ramp, and Fraction of Hours Near Peak are additionally used to construct the composite fragility score (see Section 3). Ramp Standard Deviation and Daily Load Range are used in the PCA used for clustering but not the fragility scores (see Section 4).

This within-BA normalization was chosen deliberately: CAISO and PJM operate at fundamentally different absolute scales, so a raw MW comparison would conflate system size with operational stress. Z-scoring puts both systems on a common scale where a value of +1.0 means "one standard deviation above that system's own average," making cross-BA comparisons meaningful without assuming the two grids are structurally identical.

The composite fragility score is the equal-weighted sum of the three z-scores. Equal weighting is an explicit design assumption — no single component was assumed a priori to dominate stress. The resulting composite (fragility_z) was then min-max scaled within each BA to produce a 0–1 index (fragility_01), where 0 represents the least stressed observed day and 1 the most. Because both systems operate within a comparable z-score range (roughly ± 4 to ± 6), this scaling does not distort cross-BA comparisons (Figure A1).

Across the full year, both systems' fragility scores follow a broadly similar distribution — roughly symmetric around zero, spanning approximately -4 to +6 standard deviations. Neither system shows a systematic skew toward higher fragility, consistent with the finding that they operate at comparable levels of relative stress over the study period (Figure A2).

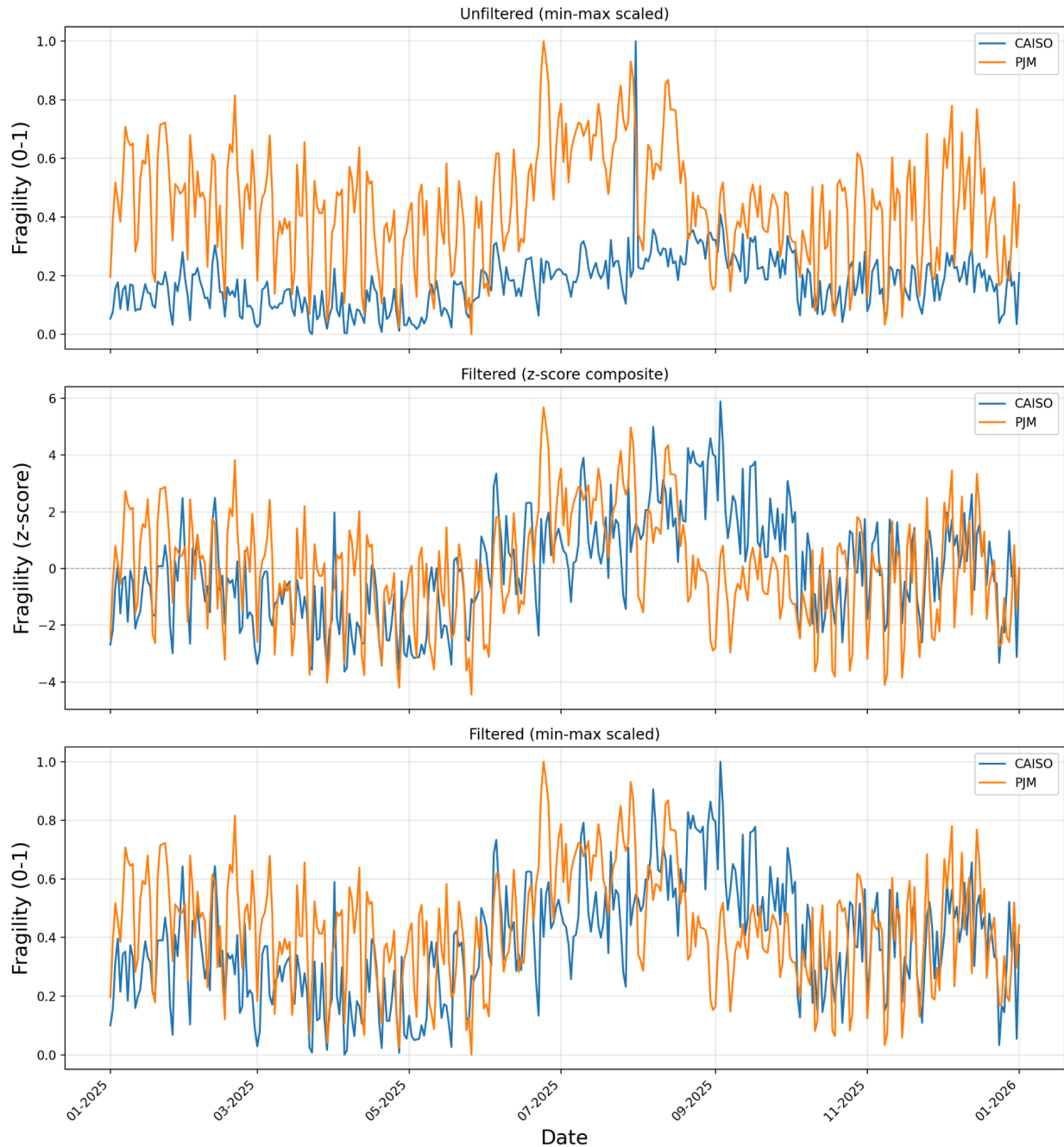


Figure A1. Unfiltered (top), filtered (middle) and normalized (bottom) time series data

Three-panel comparison of daily fragility for CAISO and PJM over calendar year 2025. Top: unfiltered min-max scaled fragility, prior to data quality correction - the CAISO spike to 1.0 in late July is a reporting artifact that compresses all other days toward zero. Middle: filtered min-max scaled fragility after spike correction, used throughout the main report. Bottom: filtered composite z-score, preserving cross-BA magnitude for reference - both systems operate within comparable range, confirming that min-max scaling does not distort cross-system comparisons.

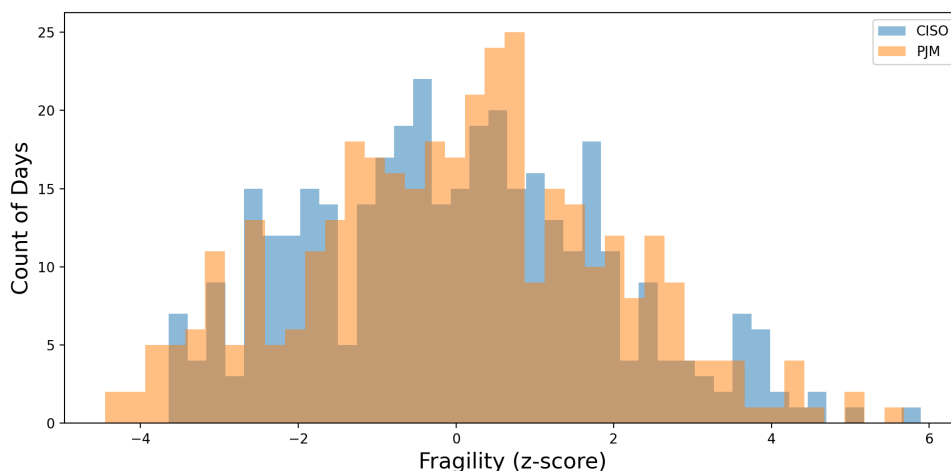


Figure A2. Distribution of daily fragility scores for both BAs

Fairly symmetric, mild right skew, upper tails reaching ~5–6 z-score. Distributions overlap substantially - differences between systems are structural, not in aggregate fragility levels.

Appendix B: Operational Regime Clustering and Validation

Daily operational regimes were identified using K-Means clustering ($k=4$) applied independently to each BA. Features were first reduced using PCA before clustering to remove collinearity across the five input features (peak load, max ramp, ramp standard deviation, daily range, and fraction of hours near peak). Three principal components were retained, accounting for 97.2% of variance in CAISO and 94.2% in PJM. Clusters were assigned post-hoc labels based on mean fragility_z. It is worth noting that the clusters do not separate cleanly along any single axis in peak load vs. ramp space — there is meaningful overlap between regimes, particularly in the moderate range. This suggests that peak load and max ramp alone do not fully explain regime membership, and other features captured by PCA (ramp variability, load persistence) contribute to cluster boundaries (see Appendix C for k selection rationale and PCA variance detail).

CAISO's high-stress regime accounts for only 43 days (12% of the year) but produces a mean fragility_z of 3.0, driven by simultaneous elevation in both peak load and ramp. PJM's most distinctive regime is the Ramp-Volatile cluster — 138 days where ramp stress is modestly elevated despite near-average load, with no clear CAISO analog. Notably, CAISO's two moderate clusters (Mild and Moderate) carry nearly identical mean fragility scores (0.61 and 0.65) despite sitting in distinct regions of feature space, suggesting the clustering is resolving operational differences that the composite score alone does not fully capture. PJM's regime structure is more evenly distributed — no single non-baseline cluster dominates the year — whereas CAISO concentrates its stress into a small, acute high-stress window. Both systems' baseline regimes are tight and well-separated from the rest, indicating that low-stress days are consistently low-stress rather than just averaging out mixed signals (Table B1).

CAISO Regimes (366 days)

Label	Cluster	Days	Mean Peak z	Mean Ramp z	Mean Fragility z
Baseline	2	128	-0.73	-0.69	-1.97
Mild	0	101	-0.42	-0.26	+0.61
Moderate	3	94	+0.51	+0.45	+0.65
High Stress	1	43	+2.04	+1.68	+3.00

PJM Regimes (366 days)

Label	Cluster	Days	Mean Peak z	Mean Ramp z	Mean Fragility z
Baseline	2	75	-0.77	-1.41	-2.60
Ramp-Volatile	1	138	-0.19	+0.27	-0.39
Moderate	0	78	-0.40	-0.13	+0.95
High Stress	3	75	+1.54	+1.05	+2.34

Table B1. Operational regime characteristics for CAISO and PJM

Derived from K-Means clustering (k=4) on PCA-reduced daily features. Clusters are ordered by mean fragility z-score within each BA. Mean peak z and mean ramp z reflect the average standardized values of daily peak load and maximum hourly ramp across all days in regime.

Feature space plots (Figure 4) show that while both high-stress regimes combine peak load and ramp stress, cluster boundaries in the moderate range are not clean - different regimes overlap meaningfully, particularly for CAISO clusters 0 and 3, which carry nearly identical mean fragility scores (0.61 and 0.65 respectively) despite occupying distinct regions of feature space. This overlap reflects the limits of a two-dimensional visualization: PCA incorporated five features, and the third and fourth components - capturing ramp standard deviation and load persistence - are driving separation that is not visible in the peak load vs. max ramp projection.

The feature contribution percentages cited in Section 4 (42/43/15 for CAISO, 45/34/21 for PJM) are derived from the relative z-score magnitudes on high-fragility days (fragility_z > 1.0) rather than from cluster centroids directly, and represent the share each component contributes to the composite score on those days (Figure 3). Pearson correlation analysis across all 366 days corroborates this structure (Figure B1): max ramp is the strongest individual predictor of fragility in both systems, followed closely by peak load. Near-peak persistence is a weaker but meaningful contributor, and notably more correlated with fragility in PJM than in CAISO. Cluster day counts and distribution across both BAs are shown in Figure B2.

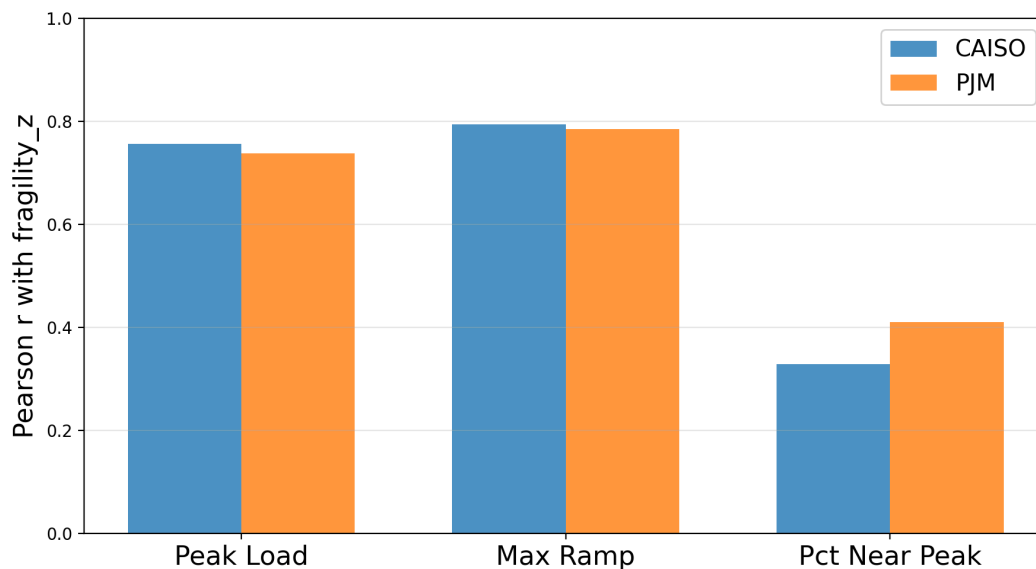


Figure B1. Pearson correlation box plot

Pearson correlation between each individual feature and the composite fragility z-score, computed across all 366 days for each BA. Max ramp is the strongest single predictor of fragility in both systems, followed closely by peak load. Near-peak persistence is a weaker but consistent contributor, more so in PJM than CAISO.

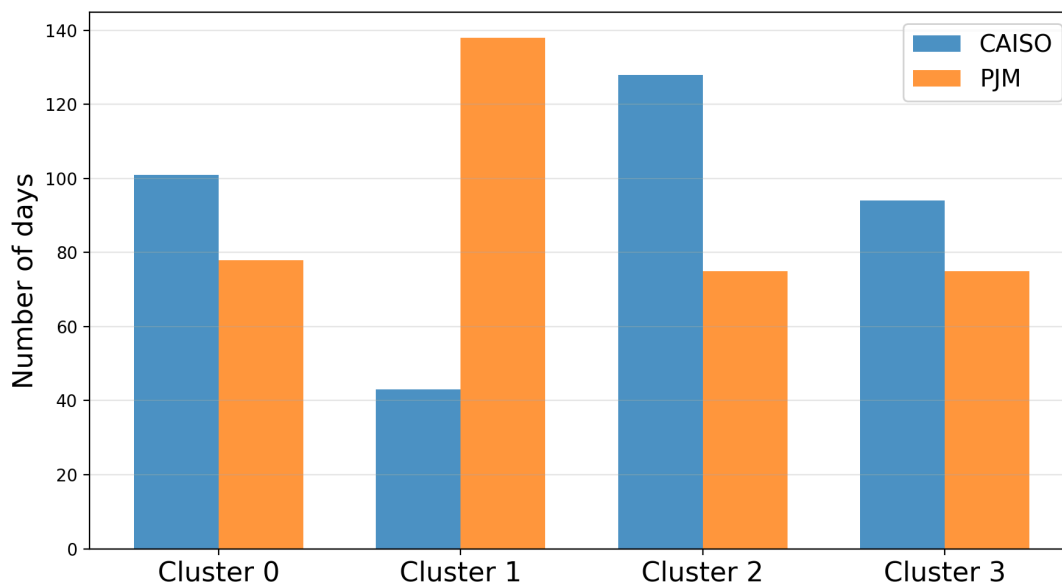


Figure B2. Cluster count bar chart

Number of days assigned to each cluster for CAISO and PJM. Clustering was performed independently per BA — cluster numbers are not comparable across systems.

Appendix C: Supplementary Notes

C.1 - Equal Feature Weighting

The composite fragility score weights peak load, max ramp, and near-peak persistence equally. This is a design assumption, not an empirically derived result. A weighted scheme could produce a more defensible composite, but would require additional external validation data.

C.2 - Choice of k=4

Four clusters were selected based on interpretability - each BA produces four regimes that are meaningfully distinct in both operational character and fragility level. Values of k=3 produce regimes that are too coarse to distinguish moderate stress from high stress; k=5 begins to split regimes that are operationally similar, adding granularity without adding insight - confirmed by the silhouette scores (Table C.1).

K	CAISO	PJM
3	0.338	0.337
4	0.346	0.387
5	0.337	0.354

Table C.1 - Silhouette Analysis
K = 4 optimizes the scores

C.3 - PCA Explained Variance

Three principal components were retained for clustering, which account for 97.2% of variance in CAISO and 94.2% in PJM, meaning the dimensionality reduction discards very little information. The first component alone captures 69.4% (CAISO) and 64.8% (PJM) of variance, reflecting the high collinearity between peak load, load range, and ramp features on high-demand days.

C.4 - Solar generation in each BA

In 2025, solar accounted for 26.6% of CAISO's annual net generation versus 2.8% for PJM - which contributes directly to the different ramp volatilities through the solar duck curve effect.¹

C.5 - One Year of Data

The study covers 366 days — enough to characterize seasonal structure and identify repeatable regime patterns, but insufficient to assess inter-year variability. Grid composition is changing rapidly: increasing solar penetration, electrification of heating and transportation, and large new loads like data centers will shift the baseline. Regimes may not generalize without retraining.

C.6 - Unsupervised Learning

The ML methods used, PCA and K-Means, are unsupervised. The regime labels are derived from the data, not compared to an external standard. Predictive modeling - forecasting which regime tomorrow will fall into - is the logical next step, and the regime labels produced here are clean targets for that work. Predictive modeling can then be used to make operational and design decisions for the grid, such as how many new data centers to build and where to build them. But a future credible forecasting model will require either a longer historical dataset or external covariates (weather forecasts, prior-day conditions).

¹ U.S. Energy Information Administration. EIA-930 Hourly Electric Grid Monitor. Washington, DC: EIA, 2025. Available at: <https://www.eia.gov/electricity/gridmonitor/>